Day 4: Hands-on Subnetting

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Course: Diploma in Cyber Security

Topic: Day 4: Hands-on Subnetting

Introduction

Subnetting means dividing one larger network into smaller networks. It helps organise devices and can reduce unnecessary network traffic.

For 192.168.10.0/24 divided into four equal parts, two host bits are borrowed, giving /26 subnets.

Task 1

Subnet	Network Address	Broadcast Address
1	192.168.10.0/26	192.168.10.63
2	192.168.10.64/26	192.168.10.127
3	192.168.10.128/26	192.168.10.191
4	192.168.10.192/26	192.168.10.255

Task 2

Device	Assigned IP
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile	192.168.10.5

Subnet: 192.168.10.0/26. Usable range is 192.168.10.1–192.168.10.62.

Task 3

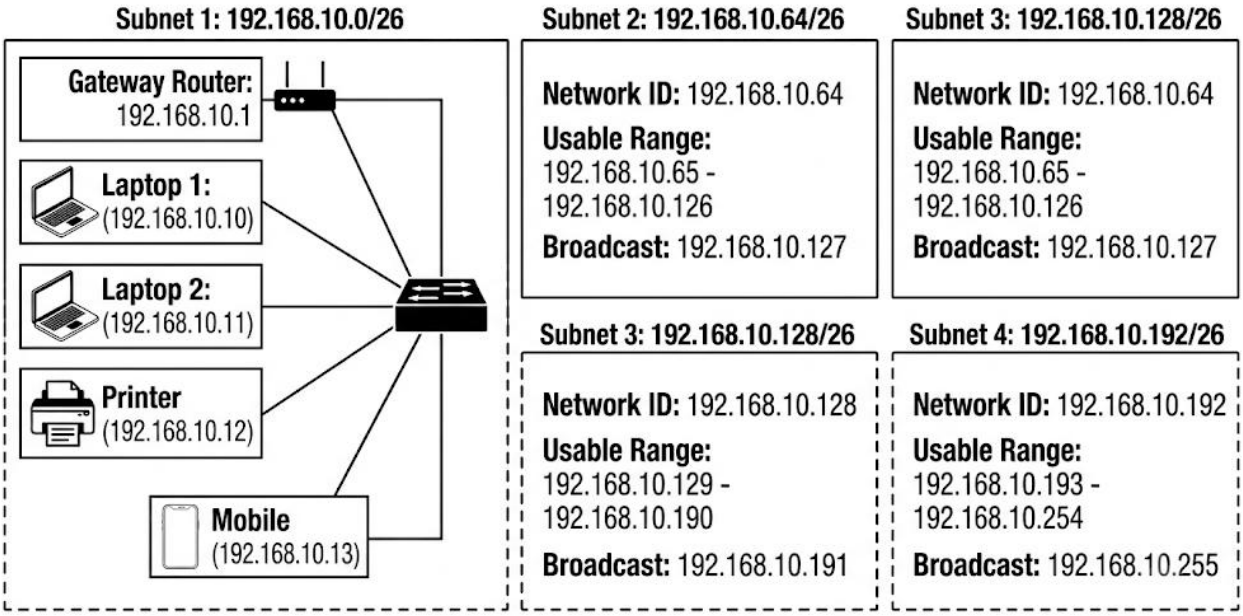


Figure: Complete IPv4 Subnetting Topology for 192.168.10.0/24 Divided into Four /26 Subnets

Draw four subnet boxes:

- Subnet 1: 192.168.10.0/26
- Subnet 2: 192.168.10.64/26
- Subnet 3: 192.168.10.128/26
- Subnet 4: 192.168.10.192/26

Inside Subnet 1, show the router, two laptops, printer and mobile with their assigned IPs.

Task 4

Step 1: Start with 192.168.10.0/24.
Step 2: Four subnets are needed, so borrow 2 host bits because $2^2 = 4$.
Step 3: The new prefix becomes /26.
Step 4: /26 gives 64 addresses per subnet.
Step 5: The block size is 64, so subnet addresses increase by 64.
Step 6: The four network addresses are .0, .64, .128 and .192.
Step 7: For each subnet, the first address is the network address and the last is the broadcast address.
Step 8: Assign device addresses only from the usable host range.

Conclusion

The subnet plan divides one /24 network into four equal /26 networks. Each subnet has enough usable addresses for a small group of devices, and the network and broadcast addresses are kept reserved.

Quick Revision

- Four equal subnets from /24 become /26.
- Each /26 has 64 total addresses.
- Each /26 has 62 usable hosts.
- Subnet blocks increase by 64.